

Year 10
Pathway 2
End of Term 1 Test

Name: SOLUTIONS

Total Marks: 64 marks
Total Time: 55 minutes

Full working out must be shown to get full marks.

Attempt all questions

Resources allowed:

1 A4 page, (1 side) of hand written notes

Calculator

1) Complete the following [7 Marks]

- a) Write $6 \times 6 \times 6$ in index notation 6^3
- b) Evaluate $21^0 = 1$
- c) Simplify $x^9 \div x^3$ x^6
- d) Simplify $(y^5)^2$ y^{10}
- e) Simplify $2w + 7x - w + 5x$ $w + 12x$
- f) Evaluate 2^5 $2 \times 2 \times 2 \times 2 \times 2 = 32$
- g) Simplify $3^5 \times 3$ (leaving your answer in index form) $= 3^6$

2) Expand the following: (simplify when necessary) [1,1,2,2,marks]

a) $4(a + 3)$

$4a + 12$

b) $5x(x + 3)$

$5x^2 + 15x$

a) $2(a + 2) + 3(2a + 1)$

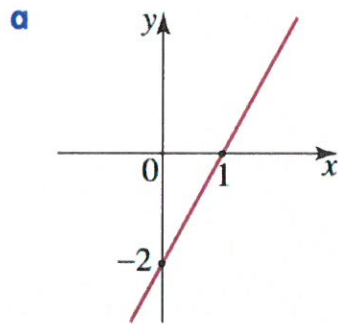
$2a + 4 + 6a + 3$
 $8a + 7$

b) $4a(3 + 2a) - 2(a + b)$

$12a + 8a^2 - 2a - 2b$
 $8a^2 + 10a - 2b$

3) Find the equations of the straight lines in the following graphs.

[3, 3 Marks]



Gradient-

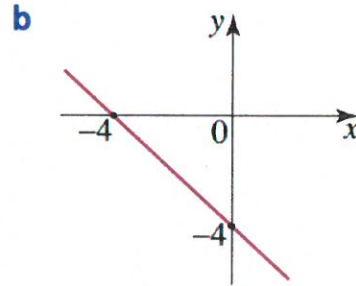
2

Y intercept-

-2

Equation-

$$y = 2x - 2$$



Gradient-

-1

Y intercept-

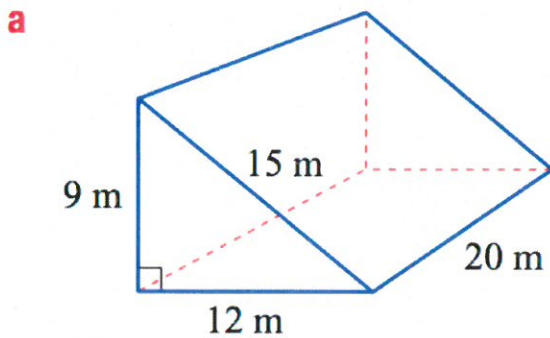
-4

Equation-

$$y = -x - 4$$

4) Calculate the surface area of the following Shapes.

[3, 3, 3- Marks]



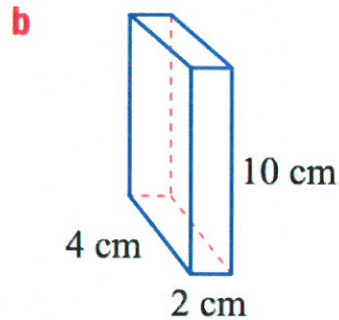
$$A = \frac{1}{2} b \times h = \frac{1}{2} (9) \times 12 \times 2 = 108$$

$$A = L \times W = 15 \times 20 = 300$$

$$= 12 \times 20 = 240$$

$$= 9 \times 20 = 180$$

$$\underline{828 \text{ m}^2}$$



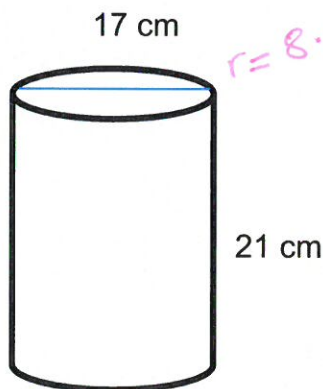
$$A = L \times W = 4 \times 10 = 40 \times 2 = 80$$

$$= 4 \times 2 = 8 \times 10 = 80$$

$$= 2 \times 10 = 20 \times 2 = 40$$

$$\underline{196 \text{ cm}^2}$$

c)



$$r = 8.5$$

$$A = \pi r^2$$

$$= \pi (8.5)^2$$

$$= 226.98$$

$$\times 2$$

$$= 453.96 \text{ cm}^2$$

$$C = 2\pi r$$

$$= 2\pi (8.5)$$

$$= 53.41 \times 21 = 1121.54 \text{ cm}^2$$

$$\boxed{SA = 1575.51 \text{ cm}^2}$$

5) Simplify the following algebraic fractions

[1, 1, 2, 2, 2, 2-marks]

a) $\frac{4a}{2}$ $2a$

b) $\frac{2a}{10} + \frac{5a}{10}$ $\frac{7a}{10}$

c) $\frac{2c}{5} + \frac{4c}{10}$ $\frac{2(2c) + 4c}{10}$
 $= \frac{8c}{10} = \frac{4c}{5}$

d) $\frac{9z}{10} - \frac{3z}{5}$
 $\frac{9z}{10} - \frac{6z}{10} = \frac{3z}{10}$

e) $\frac{6b}{3} \times \frac{4b}{10}$ $\frac{24b^2}{30} = \frac{12b^2}{15}$
 $= \frac{4b^2}{5}$

f) $\frac{5d}{6} \div \frac{5d}{8}$
 $\frac{5d}{6} \times \frac{8}{5d} = \frac{8}{6} = 1\frac{1}{3}$

6) Simplify each of the following and express your answer with positive indices.

[2, 2-marks]

a) $2a^{-5}b^2 \times 4a^{-6}b^{-4}$

$8a^{-11}b^{-2}$
 $= \frac{8}{a^{11}b^2}$

b) $(2m^{-3}n^2)^{-4}$

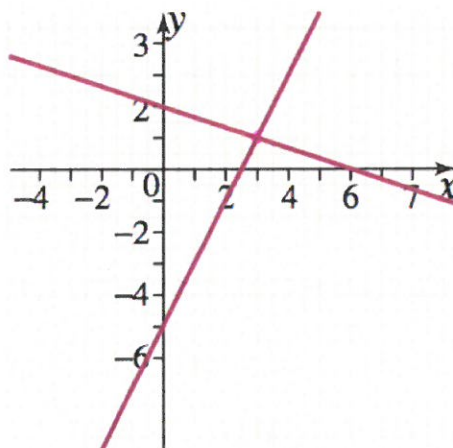
$2^{-4}m^{+12}n^{-8}$
 $\frac{m^{12}}{2^4n^8}$

7) Use the graphs below, showing the given simultaneous equations, to write the point of intersection of the graphs and, hence, the solution of the simultaneous equations.

[2 marks]

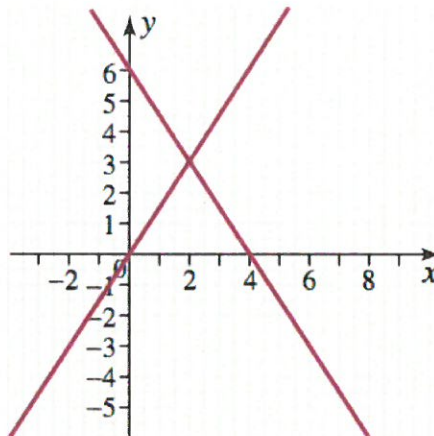
a $x + 3y = 6$
 $y = 2x - 5$

$(3, 1)$



b $3x + 2y = 12$
 $2y = 3x$

$(2, 3)$



8) Find the x- and y-intercepts of the following straight lines.

[2, 2, 2-marks]

a) $y = -12x + 6$ y int= x int=

b) $y = \frac{1}{2}x - 5$ y int= x int=

c) $y = 2x - 3$ y int= x int=

9) A fun park charges a \$12.50 entry fee and an additional \$2.50 per ride. –

[3, 2, 1, 4, Marks]

$\$ = \$2.50r + \$12.50$

a) Complete the following table of values relating the total cost to the number of rides.

Number of rides	0	2	4	6	8	10
Cost (\$)	12.50	17.50	22.50	27.50	32.50	37.50

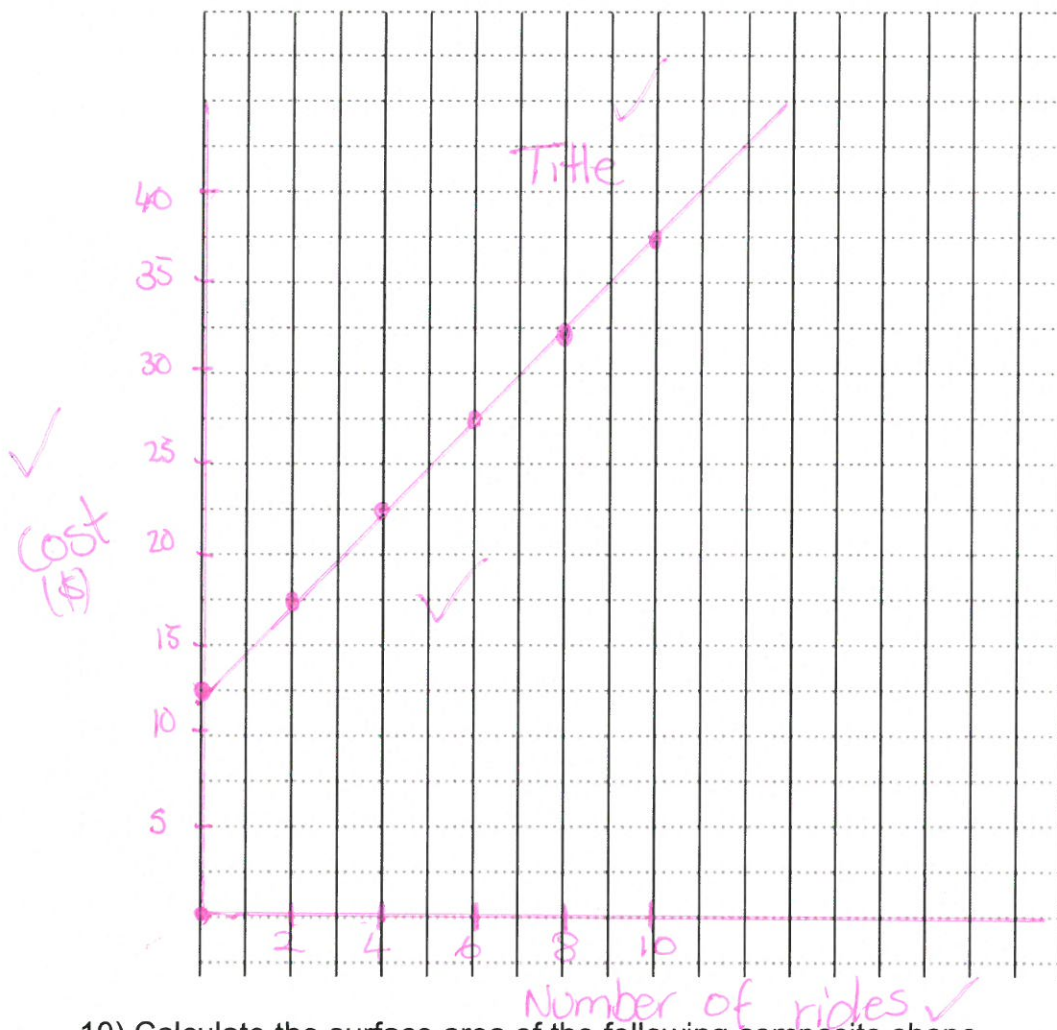
b) Find a linear equation relating total cost to the number of rides.

$\text{Cost}(\$) = \$2.50r + \$12.50$

c) Using algebra, calculate the cost for 7 rides.

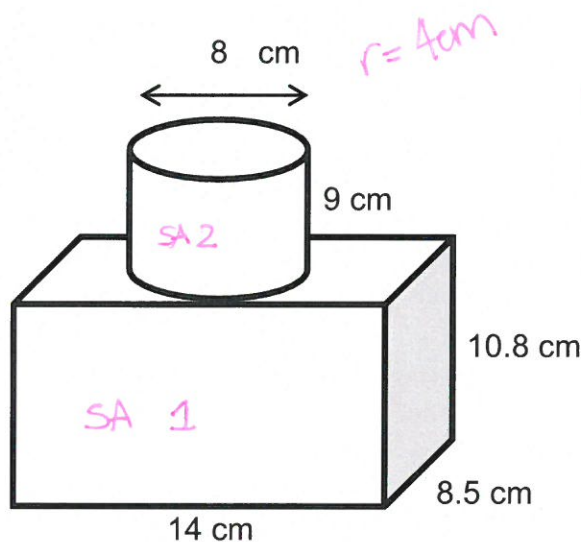
$\text{cost}(\$) = \$2.50(7) + \$12.50$
 $= \$30$

d) Sketch the linear equation on a Cartesian plane over a suitable domain.



10) Calculate the surface area of the following composite shape

[5 Marks]



$$\begin{aligned}
 SA_2 &= \pi r^2 = \pi (4)^2 \\
 &= 50.265 \text{ cm}^2 \\
 2\pi r \times h &= 2(\pi)(4) \times 9 \\
 &= 261.19 \text{ cm}^2 \\
 SA_2 &= 276.46 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 SA_1 &= L \times W = 14 \times 10.8 = 151.2 \times 2 = 302.4 \text{ cm}^2 \\
 L \times W &= 8.5 \times 10.8 = 91.8 \times 2 = 183.6 \text{ cm}^2 \\
 L \times W &= 14 \times 8.5 = 119 \times 2 = 238 \text{ cm}^2 \\
 SA &= 724 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Total SA} &= 724 + 276.46 - 50.265 \\
 &= 950.19 \text{ cm}^2
 \end{aligned}$$

End of Test

